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1. Vector and matrix operations in MATLAB

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Calculator

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Recitation due Feb 14, 2024 20:30 IST

Vector and matrix operations (External resource)

(1.0 points possible)

Array operations

- In general, MATLAB is quite intuitive when it comes to operating on matrices ar vectors or matrices **A** and **B** to form a third matrix **C** using the following comr

`C = A+B`

- If **A** is an $m \times n$ matrix and **B** is an $n \times p$ matrix, then we can calculate the prc following command:

`D = A*B`

- If $\mathbf{v}_1$ and $\mathbf{v}_2$ are two vectors of the same length, we can calculate the dot produ

`a = dot(v1,v2)`

- The transpose of a real valued matrix/vector **A** is a new matrix, $\mathbf{B} = \mathbf{A}^T$, such **A**. It is calculated using an apostrophe:

`B = A'`

Note that if the entries of **A** are complex, then the apostrophe in MATLAB creat just the transpose.

- The final MATLAB operation we will look at in this section is "element-wise" mu **A** and **B** of the same size. Then the element-wize product takes each entry in entry in **B**

e.g. if

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$$

Then the element-wise product is

$$\mathbf{C} = \mathbf{A} \circ \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 6 & 3 \end{bmatrix}$$

The matlab command for this operation is

`C = A.*B`

Define the following matrices and vectors:

$$\mathbf{X} = \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}$$
$$\mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

Use MATLAB's built-in functions to define the following new arrays:

1. $\mathbf{A} = \mathbf{XY}$
2. $\mathbf{B} = \mathbf{YX}$
3. $\mathbf{C} = \frac{1}{2}[\mathbf{X} - \mathbf{Y}^T]$
4. $\mathbf{D} = \mathbf{X} \circ \mathbf{Y}$
5. $a = \mathbf{u} \cdot \mathbf{v}$
6. $b = \mathbf{v}^T \mathbf{X} \mathbf{u}$

Script ?

 Save  Reset

 [MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

1 | % We provide definitions of X, Y, u and v for you

 Calculator

```
2 X = [2 1; 0 4];
3 Y = [3 4; 1 1];
4 u = [-1;1];
5 v = [4;-3];
6
7 A = X*Y;
8 B = Y*X;
9 C = 0.5*(X-Y');
10 D = X.*Y;
11 a = u'*v;
12 b = v'*X*u;
```

▶ Run Script ?

Assessment: All Tests Passed Submit ?

- ✔ Value of A
- ✔ Value of B
- ✔ Value of C
- ✔ Value of D
- ✔ Value of a

1. Vector and matrix operations in MATLAB Hide Discussion

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